

Python - Tuples

Tuple is one of the built-in data types in Python. A Python tuple is a sequence of comma separated items, enclosed in parentheses (). The items in a Python tuple need not be of same data type.

Following are some examples of Python tuples –

```
tup1 = ("Rohan", "Physics", 21, 69.75)
tup2 = (1, 2, 3, 4, 5)
tup3 = ("a", "b", "c", "d")
tup4 = (25.50, True, -55, 1+2j)
```

The empty tuple is written as two parentheses containing nothing –

```
tup1 = ();
```

To write a tuple containing a single value you have to include a comma, even though there is only one value –

```
tup1 = (50,);
```

Following are the points to be noted –

- In Python, tuple is a sequence data type. It is an ordered collection of items. Each item in the tuple has a unique position index, starting from 0.
- In C/C++/Java array, the array elements must be of same type. On the other hand, Python tuple may have objects of different data types.
- Python tuple and list both are sequences. One major difference between the two is, Python list is mutable, whereas tuple is immutable. Although any item from

the tuple can be accessed using its index, and cannot be modified, removed or added.

Accessing Values in Tuples

To access values in tuple, use the square brackets for slicing along with the index or indices to obtain value available at that index. For example –

```
tup1 = ('physics', 'chemistry', 1997, 2000);
tup2 = (1, 2, 3, 4, 5, 6, 7 );
print ("tup1[0]: ", tup1[0]);
print ("tup2[1:5]: ", tup2[1:5]);
```

When the above code is executed, it produces the following result –

```
tup1[0]:  physics
tup2[1:5]:  [2, 3, 4, 5]
```

Updating Tuples

Tuples are immutable which means you cannot update or change the values of tuple elements. You are able to take portions of existing tuples to create new tuples as the following example demonstrates –

```
tup1 = (12, 34.56);
tup2 = ('abc', 'xyz');

# Following action is not valid for tuples
# tup1[0] = 100;

# So let's create a new tuple as follows
tup3 = tup1 + tup2;
print (tup3);
```

When the above code is executed, it produces the following result –

```
(12, 34.56, 'abc', 'xyz')
```

Delete Tuple Elements

Removing individual tuple elements is not possible. There is, of course, nothing wrong with putting together another tuple with the undesired elements discarded.

To explicitly remove an entire tuple, just use the **del** statement. For example –

```
tup = ('physics', 'chemistry', 1997, 2000);
print (tup);
del tup;
print ("After deleting tup : ");
print (tup);
```

This produces the following result. Note an exception raised, this is because after **del tup** tuple does not exist any more –

```
('physics', 'chemistry', 1997, 2000)
After deleting tup :
Traceback (most recent call last):
  File "test.py", line 9, in <module>
    print (tup);
NameError: name 'tup' is not defined
```

Python Tuple Operations

In Python, Tuple is a sequence. Hence, we can concatenate two tuples with + operator and concatenate multiple copies of a tuple with "*" operator. The membership operators "in" and "not in" work with tuple object.

Python Expression	Results	Description
(1, 2, 3) + (4, 5, 6)	(1, 2, 3, 4, 5, 6)	Concatenation
('Hi!') * 4	('Hi!', 'Hi!', 'Hi!', 'Hi!')	Repetition
3 in (1, 2, 3)	True	Membership

Even if there is only one object in a tuple, you must give a comma after it. Otherwise, it is treated as a string.

Indexing, Slicing, and Matrixes

Because tuples are sequences, indexing and slicing work the same way for tuples as they do for strings. Assuming following input –

```
L = ('spam', 'Spam', 'SPAM!')
```

Python Expression	Results	Description
L[2]	'SPAM!'	Offsets start at zero
L[-2]	'Spam'	Negative: count from the right
L[1:]	['Spam', 'SPAM!']	Slicing fetches sections

No Enclosing Delimiters

Any set of multiple objects, comma-separated, written without identifying symbols, i.e., brackets for lists, parentheses for tuples, etc., default to tuples, as indicated in these short examples –

```
print ('abc', -4.24e93, 18+6.6j, 'xyz');
x, y = 1, 2;
print ("Value of x , y : ", x,y);
```

When the above code is executed, it produces the following result –

```
abc -4.24e+93 (18+6.6j) xyz
Value of x , y : 1 2
```

Built-in Functions with Tuples

Following are the built-in functions we can use with tuples –

Sr.No.	Function with Description
1	cmp(tuple1, tuple2) Compares elements of both tuples.
2	len(tuple) Gives the total length of the tuple.
3	max(tuple) Returns item from the tuple with max value.
4	min(tuple) Returns item from the tuple with min value.
5	tuple(seq) Converts a list into tuple.

TOP TUTORIALS

[Python Tutorial](#)
[Java Tutorial](#)
[C++ Tutorial](#)
[C Programming Tutorial](#)
[C# Tutorial](#)
[PHP Tutorial](#)
[R Tutorial](#)
[HTML Tutorial](#)
[CSS Tutorial](#)
[JavaScript Tutorial](#)
[SQL Tutorial](#)

TRENDING TECHNOLOGIES

[Cloud Computing Tutorial](#)
[Amazon Web Services Tutorial](#)
[Microsoft Azure Tutorial](#)
[Git Tutorial](#)
[Ethical Hacking Tutorial](#)

[Docker Tutorial](#)
[Kubernetes Tutorial](#)
[DSA Tutorial](#)
[Spring Boot Tutorial](#)
[SDLC Tutorial](#)
[Unix Tutorial](#)

CERTIFICATIONS

[Business Analytics Certification](#)
[Java & Spring Boot Advanced Certification](#)
[Data Science Advanced Certification](#)
[Cloud Computing And DevOps](#)
[Advanced Certification In Business Analytics](#)
[Artificial Intelligence And Machine Learning](#)
[DevOps Certification](#)
[Game Development Certification](#)
[Front-End Developer Certification](#)
[AWS Certification Training](#)
[Python Programming Certification](#)



Chapters ▾

Categories ≡

[Online Java Compiler](#)
[Online Python Compiler](#)
[Online Go Compiler](#)
[Online C Compiler](#)
[Online C++ Compiler](#)
[Online C# Compiler](#)
[Online PHP Compiler](#)
[Online MATLAB Compiler](#)
[Online Bash Terminal](#)
[Online SQL Compiler](#)
[Online Html Editor](#)

[ABOUT US](#) | [OUR TEAM](#) | [CAREERS](#) | [JOBS](#) | [CONTACT US](#) | [TERMS OF USE](#) |
[PRIVACY POLICY](#) | [REFUND POLICY](#) | [COOKIES POLICY](#) | [FAQ'S](#)



Tutorials Point is a leading Ed Tech company striving to provide the best learning material on technical and non-technical subjects.

© Copyright 2026. All Rights Reserved.